Department of Physics
PCS 125–Waves and Fields

Please insert in the circle the initial of your family name

Please Print Clearly:

Student: _____ _____
Family Name *First Name*

Student Number: _____ **Student Section:** _____

Mid-Term Test – February 27, 2015

Check the name of your professor:

Dr. Chithrani		Dr. Douplik		Dr. Hu		Dr. Manzhos	Dr.Toronov

Read the following attentively

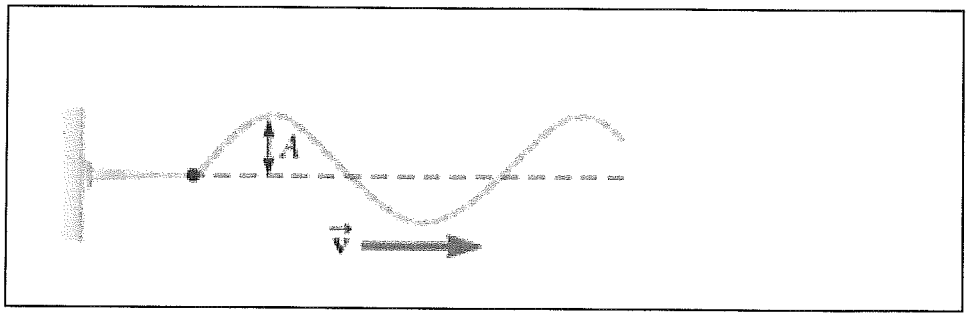
- **Duration of the test: 1 hour and 30 minutes (90 minutes).**
- **The test consists of 1 full answer question (5 marks) and 12 multiple-choice questions (1 mark each). Answer ALL questions.**
- **For each multiple choice question:**
  - **You need to show, in the space provided in this booklet, a justification for your answer. If no work or correct argument is shown, the resultant mark may be zero, even if the answer in the bubble sheet is correct.**
  - **The work shown in your exam booklet may be checked, <u>at least for one</u> of the questions, selected randomly.**
  - **<u>Transfer your choices to the bubble sheet only when you're sure of them.</u>**
  - **Some answers may be rounded off.**
- **For the full answer question, show, in the space provided, all your work. For part marks to be awarded, the steps leading to the answers in each part of the problem have to be organized, unambiguous and clear.**
- **Pagers and cell phones must be silenced and placed out of reach. Hats and earphones are not allowed. If a phone rings, its owner is responsible for disrupting the exam and the phone will be confiscated or the owner will be asked to leave the room resulting in a zero on the test.**
- **Sharing of pens, erasers and calculators is not permitted.**
- **Talking to another student or glancing over another student's paper is not permitted and may result in a charge of academic misconduct.**
- **Formula sheet provided with the test booklet. No other sheets allowed.**
- **The only calculators allowed are either the Sharp EL-546 or Casio FX-991. If you are found to have another type of calculator (even if not programmable) <u>it will be confiscated</u> for the duration of the test.**

_____

Student signature (no signature, no mark!)

FULL ANSWER QUESTION (5 marks)

The string shown in Figure is driven at a frequency of 5.00 Hz. The amplitude of the motion is A= 12.00 cm, and the wave speed is v=20 m/s. Furthermore, the wave is such that y=0 at x=0 and t=0. Determine the following:



Place your final answers to the following questions in the appropriate boxes.
Do not forget the units.

a) (1 mark) What is the angular frequency ?

0.75 0.25 (units)

$\omega = 2\pi f = 2\pi(5\ s^{-1}) = 31.4\ rad/s$

b) (1 mark) What is the wave number?

$\lambda = v/f = \frac{20}{5} = 4\ m.$

$K = \frac{2\pi}{\lambda} = \frac{2\pi}{4} = 1.57\ cm^{-1}$ → 0.25

0.75

c) (1 mark) Write an expression for the wave function

0.25 0.25 0.25

$y = A\sin(kx - \omega t + \phi) = 0.12\sin(1.57x - 31.4t)$

0.25

d) (1 mark) Calculate the maximum transverse speed of an element of the string

$y = 0.12 \sin(1.57x - 31.4t)$

$\frac{\partial y}{\partial t} = -A\omega \cos(kx - \omega t)$

$A\omega = (0.12\,m)(31.4\,rad/s)$

0.25

$= 3.77\,m/s \rightarrow 0.25$

0.5

e) (1 mark) Calculate the maximum transverse acceleration of an element of the string

$a_y = \frac{\partial v_y}{\partial t} = \frac{\partial}{\partial t} -A\omega \cos(kx - \omega t)$

$= -A\omega^2 \sin(kx - \omega t)$

max accn $= A\omega^2 = 118\,m/s^2$

0.25 0.5 0.25

MULTIPLE CHOICE QUESTIONS (each worth 1 mark)

1. For a simple harmonic oscillator with equilibrium position at x = 0
   a. The velocity is always in the same direction as the displacement from equilibrium.
   b. The velocity is always in the same direction as acceleration.
   c. The direction of the acceleration is always opposite to the direction of the displacement from equilibrium.
   d. The speed has its minimum zero value when oscillating mass passes through the equilibrium position.
   e. None of the above is true.

2. A body of mass 4.0 kg is suspended by a spring which stretches 10 cm when the mass is attached to it. It is then displaced downward an additional 8 cm and released from rest at t = 0. Its position (in meters) as a function of time is approximately
   a. $y = 0.08 \sin 9.9t$
   b. $y = 0.18 \cos 9.9t$
   c. $y = 0.10 \cos (9.9t + 0.1)$
   d. $y = 0.10 \sin (9.9t + 4.0)$
   e. $y = 0.08\cos 9.9t$

3. At sea level, at latitude where, $g = 9.80\ m/s^2$ a simple pendulum that takes 2.00 s for a complete swing back and forth has a length of 0.993 m.
   The value of g (in m/s^2) at a location where the length of such a pendulum is 0.970 m is
   a. 9.83
   b. 10.8
   c. 10.0
   d. 9.57
   e. 8.55

4. A transverse wave on a string is described by the expression $y = (0.120m)\cos(\frac{\pi}{8}x + 4\pi t)$.
   The speed of propagation of this wave is
   a. 8 m/s
   b. 0.031 m/s
   c. 2 m/s
   d. 32 m/s
   e. 0.24 m/s

5. Ali attaches a length of nylon fishing line to a post. He stretches it out and shakes the end of the rope in his hand back and forth to produce waves on the line. The most efficient way for him to decrease the wavelength is to
   a. increase the tension on the hose and shake the end more times per second.
   b. decrease the tension on the hose and shake the end more times per second.
   c. increase the tension on the hose and shake the end fewer times per second.
   d. decrease the tension on the hose and shake the end fewer times per second.
   e. keep the tension and frequency the same but decrease the length of the hose.

6. Two harmonic waves are described by

$$y_1 = (3\,\text{m})\sin\left(\frac{4}{\text{m}}x - \frac{700}{\text{s}}t\right)$$

$$y_2 = (3\,\text{m})\sin\left(\frac{4}{\text{m}}x - \frac{700}{\text{s}}t - 2\right)$$

What is the amplitude of the resultant wave?
   a. 8.0 m
   b. 4.3 m
   c. 6.0 m
   d. 3.2 m
   e. 3.0 m

7. A string is stretched and fixed at both ends, 200 cm apart. If the density of the string is 0.015 g/cm, and its tension is 600 N, what is the fundamental frequency?
   a. 316 Hz
   b. 632 Hz
   c. 158 Hz
   d. 215 Hz
   e. 79 Hz

8. A clarinet behaves like a tube closed at one end. If its length is 1.0 m, and the velocity of sound is 344 m/s, what is its fundamental frequency (in Hz)?
   a. 264
   b. 140
   c. 86
   d. 440
   e. 172

9. If a 1.00-kHz sound source moves at a speed of 50.0 m/s toward a listener who moves at a speed of 30.0 m/s in a direction away from the source, what is the apparent frequency heard by the listener?
   a. 796 Hz
   b. 949 Hz
   c. 1000 Hz
   d. 1068 Hz
   e. 1273 Hz

10. A sound wave from a police siren has an intensity of 100.0 W/m^2 at a certain point; a second sound wave from a nearby ambulance has an intensity level that is 10 dB greater than the police siren's sound wave at the same point. What is the sound level of the sound wave due to the ambulance?

   a. 150dB
   b. 85dB
   c. 234dB
   d. 21dB
   e. 182Db

11. Assume a change at the source of sound reduces the wavelength of a sound wave in air by a factor of 2. What happens to its frequency?

a. It increases by a factor of 4.
b. It increases by a factor of 2.
c. It is unchanged.
d. It decreases by a factor of 2.
e. It changes by an unpredictable factor.

12. An object moves in simple harmonic motion with amplitude of 5.00 cm.
When the speed of the object is equal to 1/3 of its maximum speed, the distance of the object from its equilibrium position x = 0 is

a. 2.35 cm
b. 4.71 cm
c. 4.48 cm
d. 3.73 cm
e. 1.67 cm

PCS125 Constants and Formulae

$I_0 = 1.00 \times 10^{-12}\mathrm{W/m^2}$ $\quad$ $c = 3 \times 10^8$ m/s

$\rho_{\mathrm{air}} = 1.2\ \mathrm{kg/m^3}$ $\quad$ $G = 6.6742 \times 10^{-11}\mathrm{N \cdot m^2 \cdot kg^{-2}}$

$g = 9.8\ \ \mathrm{m \cdot s^{-2}}$ $\quad$ $k = 8.99 \times 10^9\ \mathrm{N \cdot m^2 \cdot C^{-2}}$

$e = 1.6 \times 10^{-19}$ C $\quad$ 1 eV$=1.6\times10^{-19}$ J

$m_e = 9.1094 \times 10^{-31}$ kg $\quad$ $m_p = 1.6726 \times 10^{-27}$ kg

$\mu_0 = 4\pi \times 10^{-7}\ \mathrm{T \cdot m \cdot A^{-1}}$

$$\sin a \pm \sin b = 2\cos\left(\frac{a \mp b}{2}\right)\sin\left(\frac{a \pm b}{2}\right) \qquad \sin(a \pm b) = \sin a \cos b \pm \cos a \sin b$$

$$\omega_0 = \sqrt{\frac{k}{m}} \qquad \omega_0 = \sqrt{\frac{g}{L}} \qquad x = A\exp\left(-\frac{b}{2m}t\right)\cos(\omega t) \qquad \omega = \sqrt{\omega_0^2 - \left(\tfrac{b}{2m}\right)^2}$$

$$A = \frac{F_0/m}{\sqrt{\left(w^2-\omega_0^2\right)^2+(b\omega/m)^2}} \qquad v = 331\sqrt{1+\frac{T_c}{273}}\ \mathrm{m/s} \qquad \beta = 10\log\frac{I}{I_0}$$

$$B = \frac{-\Delta P}{\Delta V/V_i} \qquad v = \sqrt{\frac{T}{\mu}} \qquad v = \sqrt{Y/\rho} \qquad v = \sqrt{B/\rho}$$

$$s(x,t) = s_{\max}\cos(kx - \omega t) \qquad \Delta P_{\max} = \rho v \omega s_{\max} \qquad \wp = \tfrac{1}{2}\rho A \omega^2 s_{\max}^2 v$$

$$I = \frac{\wp}{A} = \frac{\wp}{4\pi r^2} \qquad I = \frac{\Delta P_{\max}^2}{2\rho v} \qquad \wp = \tfrac{1}{2}\mu\omega^2 A^2 v \qquad f' = \left(\frac{v + v_O}{v - v_S}\right) f$$

$$f_B = |f_1 - f_2| \qquad F = G\frac{m_1 m_2}{r^2} \qquad F = k\frac{q_1 q_2}{r^2} \qquad F = q\vec{v} \times \vec{B}$$

$$W_{A\to B}^{\mathrm{field}} = \int_A^B \vec{F} \cdot d\vec{r} \qquad U_A - U_B = q(V_A - V_B) = W_{A\to B}^{\mathrm{field}} \qquad U(r) = -\frac{GMm}{r}$$

$$I = nqAv_D \qquad V_H = \frac{qB}{nqt}$$

$$\vec{B} = \frac{\mu_0 I}{4\pi}\int \frac{\mathrm{d}\vec{s} \times \hat{r}}{r^2}$$

$B = \dfrac{\mu_0 I}{2\pi a}$ (at a distance a from an infinitely long wire)

$B = \dfrac{\mu_0 I\theta}{4\pi a}$ (at the centre of an arc wire of radius a subtended by an angle θ)

$B = \dfrac{\mu_0 I a^2}{2\left(a^2 + x^2\right)^{3/2}}$ (at a distance x along the centre line axis of a circular loop of wire of radius a)